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Bowden et al.

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(54) **ENGINE HAVING AN OPEN FAN WITH
REDUCED BOUNDARY LAYER INDUCED
DISTORTION**

(58) **Field of Classification Search**

CPC F01D 9/065

See application file for complete search history.

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(51) **Int. Cl.**

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F01D 25/02 (2006.01)

F01D 25/12 (2006.01)

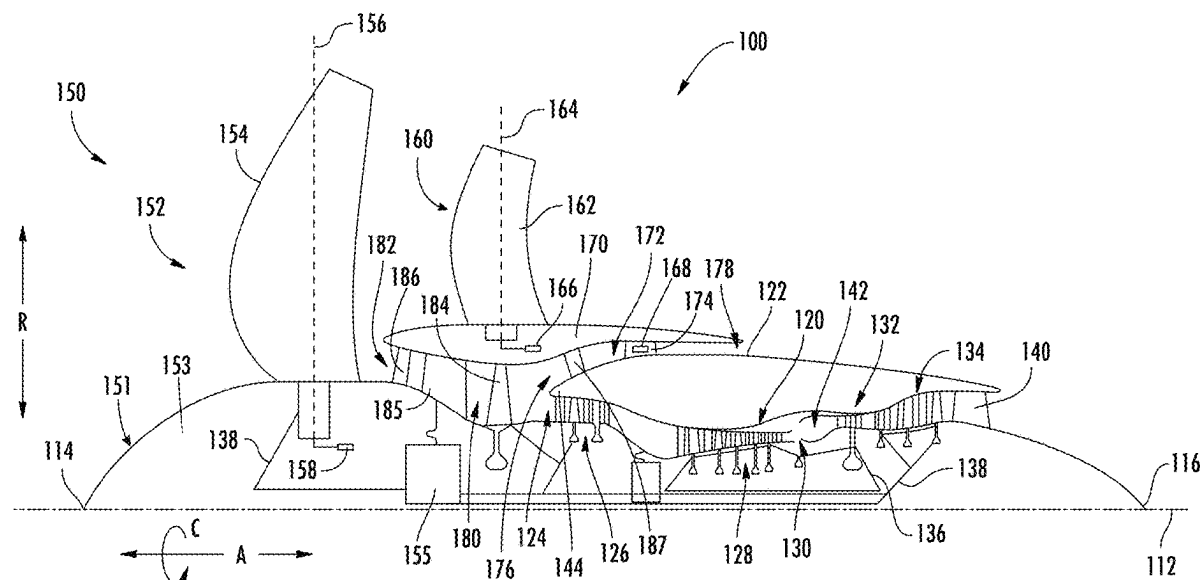
(52) **U.S. Cl.**

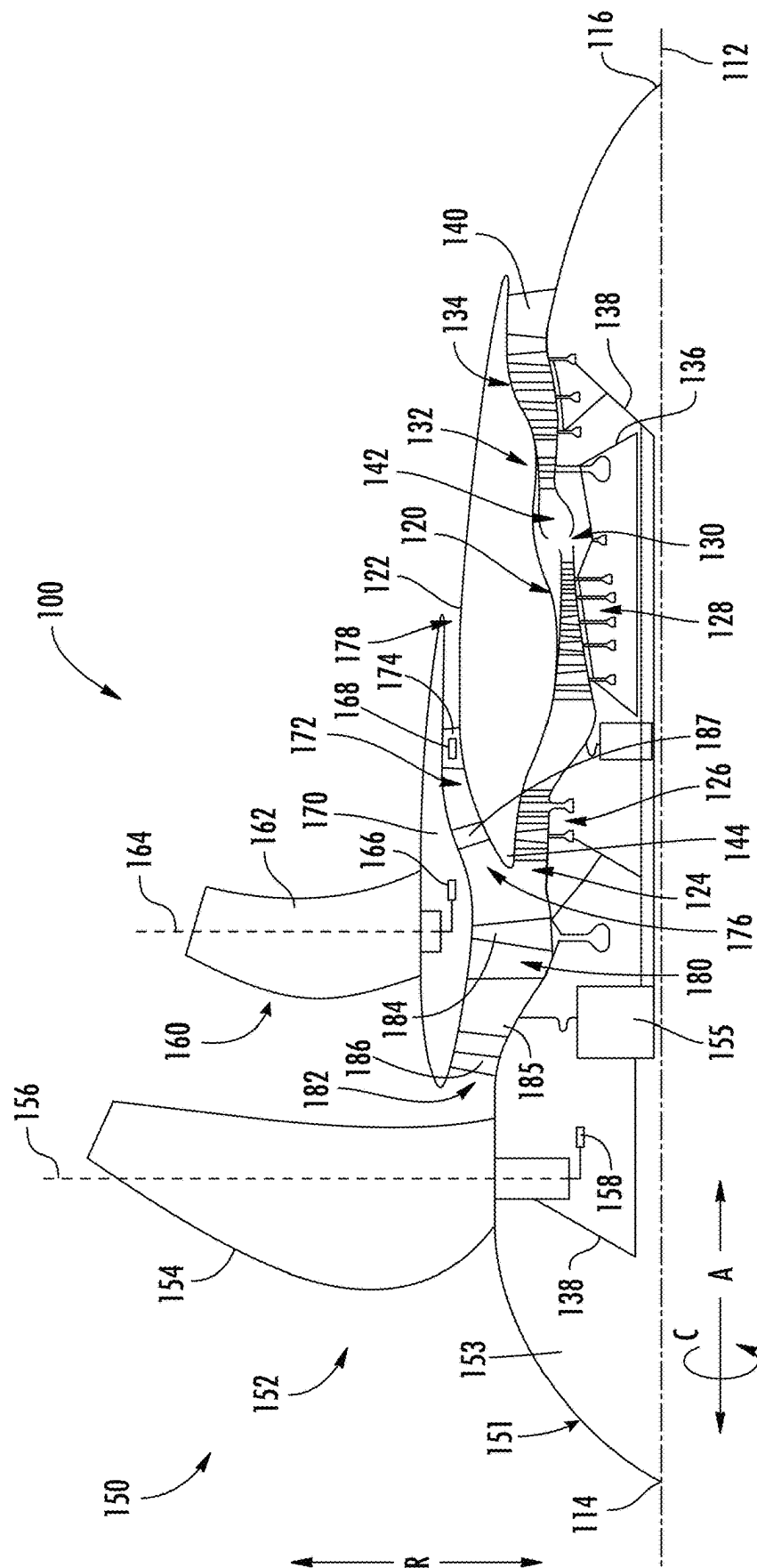
CPC **F02C 7/042** (2013.01); **F01D 25/02**
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(57) **ABSTRACT**

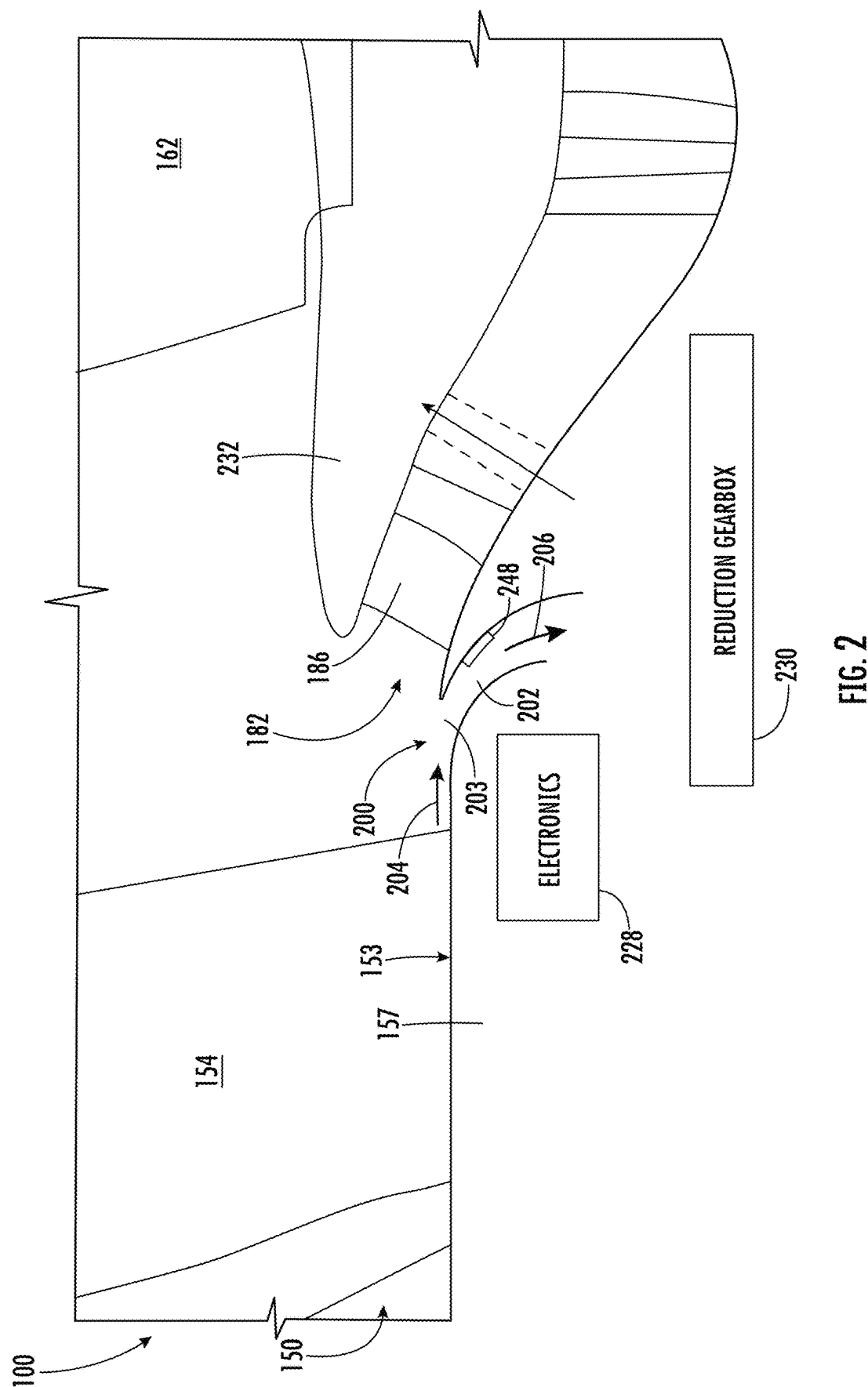
An engine includes a fan section having an open fan defining an axial centerline and a rotor. The rotor includes a plurality of fan blades mounted to a hub and being rotatable about the axial centerline. The engine also includes a turbomachine downstream of the fan section. The turbomachine includes a core cowl that defines an engine inlet and encloses turbomachinery. Further, the engine includes an airflow diverting assembly downstream of the fan section and upstream of the engine inlet. The airflow diverting assembly includes at least one fluid passageway formed into at least a portion of the hub for diverting at least a portion of a boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the turbomachinery.

8 Claims, 7 Drawing Sheets





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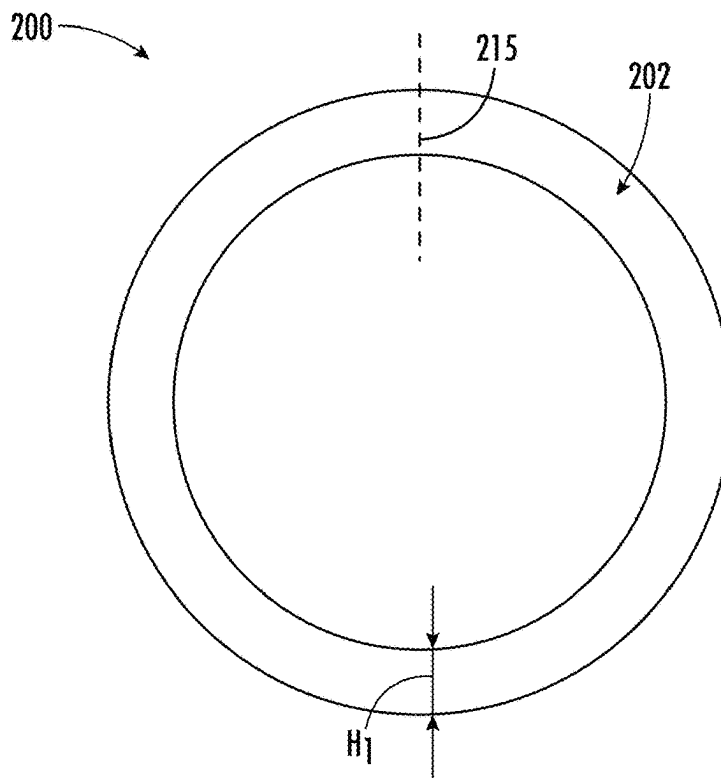


FIG. 3

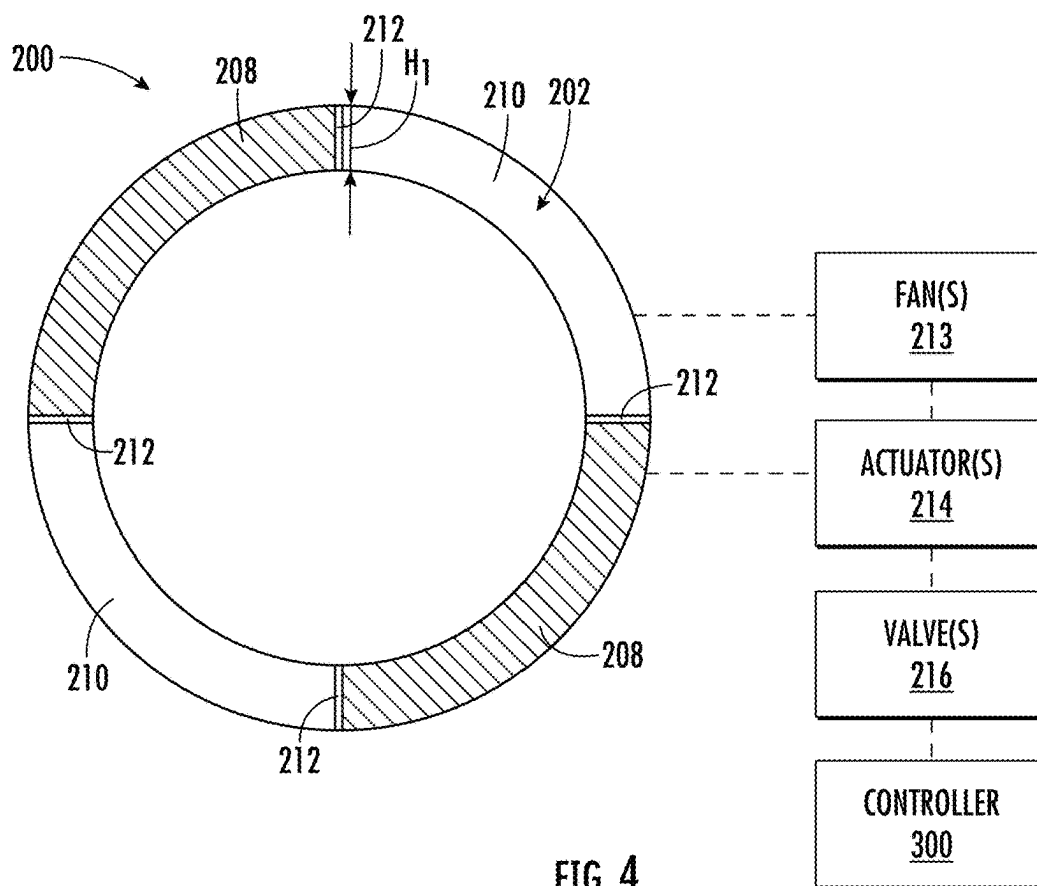
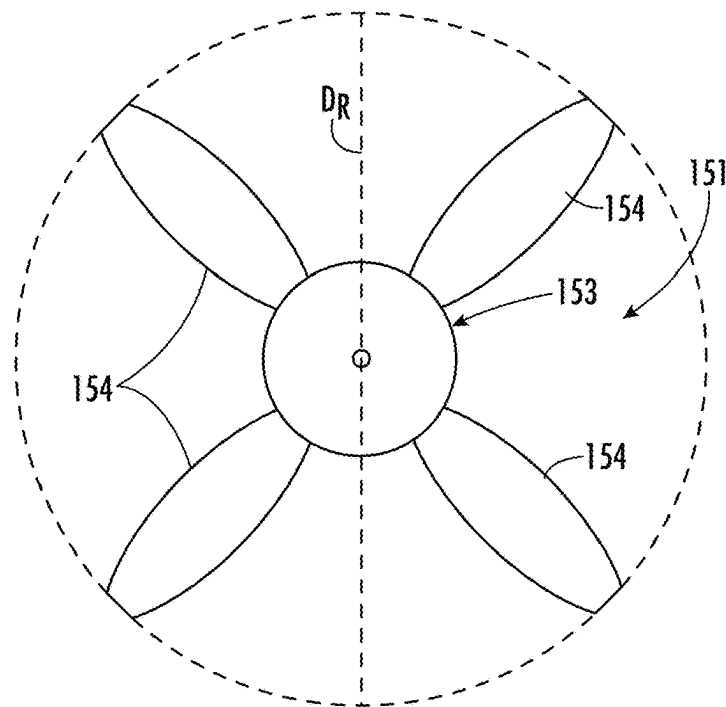
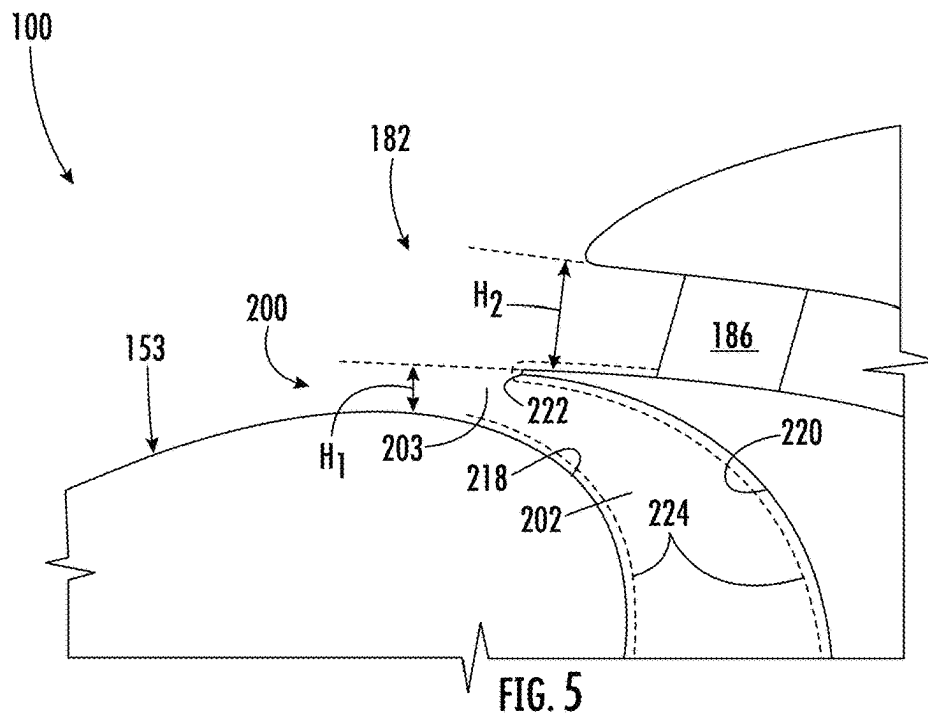


FIG. 4



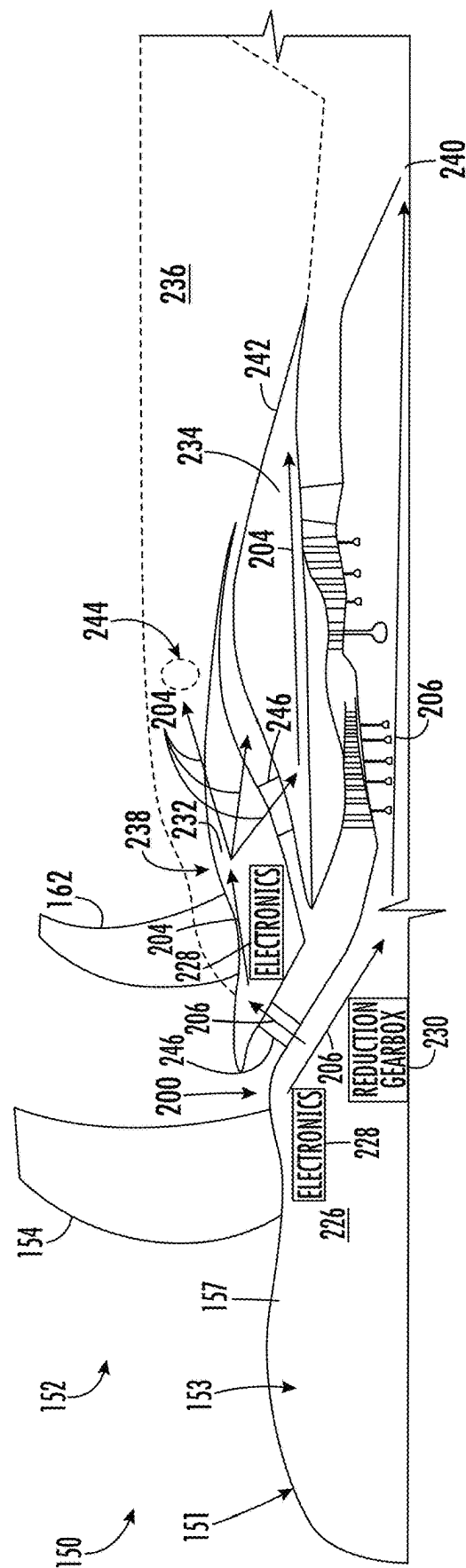


FIG. 7

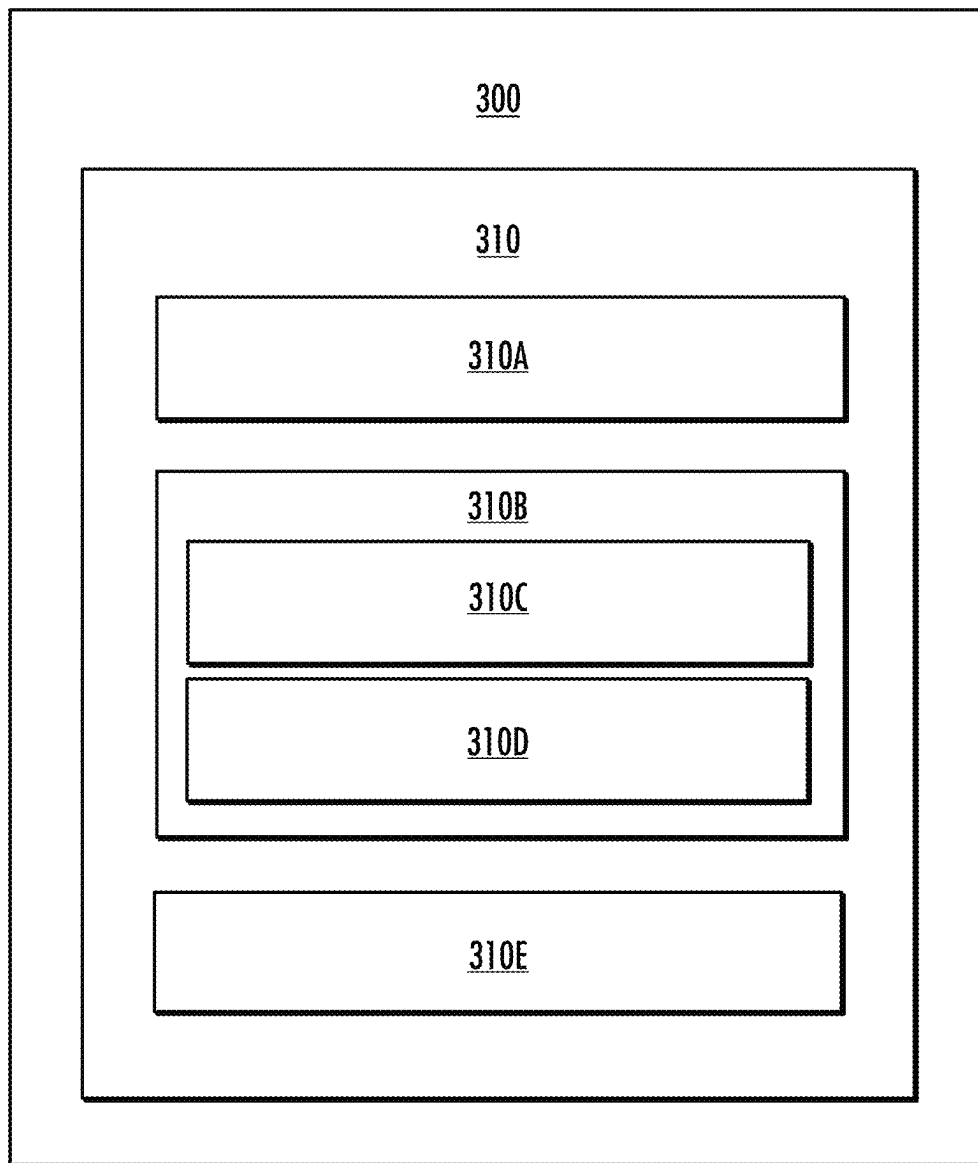


FIG. 8

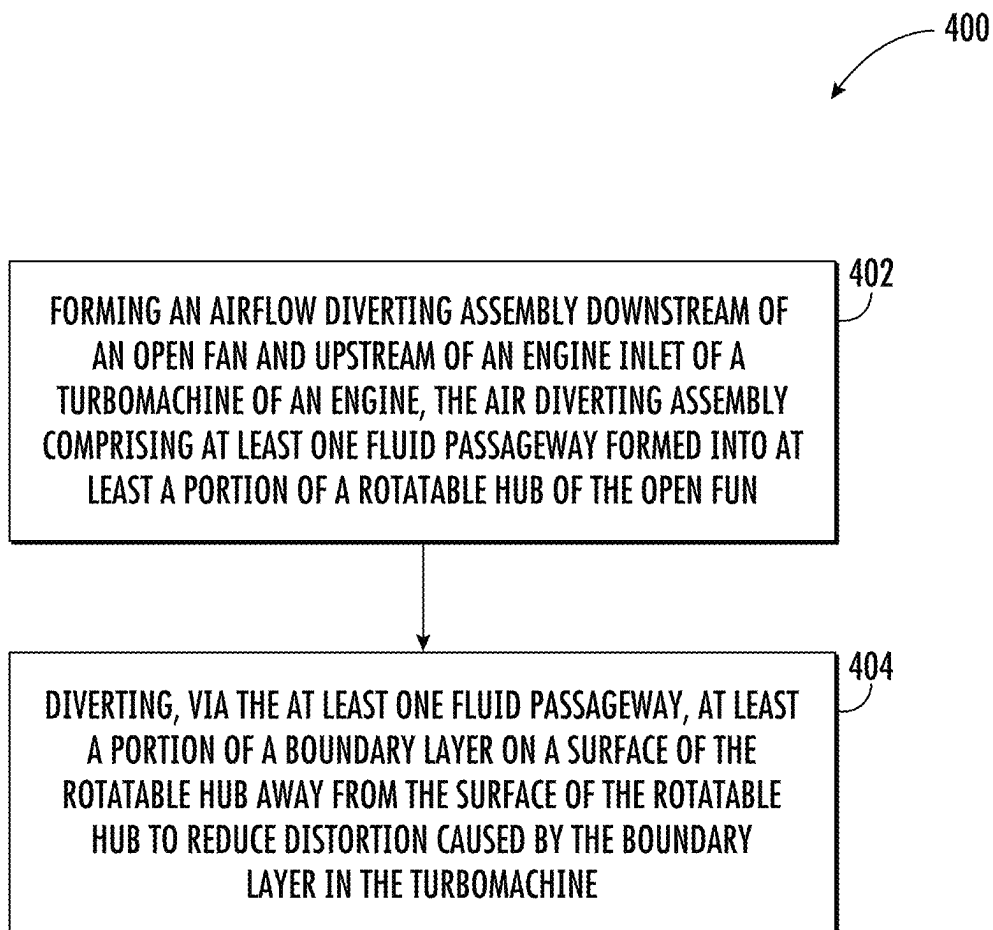


FIG. 9

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ENGINE HAVING AN OPEN FAN WITH REDUCED BOUNDARY LAYER INDUCED DISTORTION

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a divisional of and claims priority to U.S. application Ser. No. 18/171,698 entitled “Engine Having an Open Fan with Reduced Boundary Layer Induced Distortion”, filed Feb. 21, 2023, the entire disclosure of which is hereby expressly incorporated by reference herein.

FIELD

The present disclosure relates generally to turbofan engines and turboprop engines, or more particularly to an engine having an open or unducted fan with reduced boundary layer induced distortion.

BACKGROUND

A turbofan engine generally includes a turbomachine and a rotor assembly, or fan section, with a single stage of unducted rotor blades. Generally, the turbomachine includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. The turbomachine also typically includes a core cowl that defines an annular core inlet. As such, the core cowl further encloses at least in part a low pressure system and a high pressure system. For example, in some instances, the core cowl encloses and supports at least in part a booster or low pressure (“LP”) compressor for pressurizing air that enters the turbomachine through a core inlet. A high pressure (“HP”), multi-stage, axial-flow compressor receives pressurized air from the LP compressor and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor of the combustion section where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

High energy combustion products flow from the combustor downstream to an HP turbine. The HP turbine drives the HP compressor through an HP shaft. In this regard, the HP turbine is drivingly coupled with the HP compressor. The high energy combustion products then flow to an LP turbine. The LP turbine drives the LP compressor and components of the fan section through an LP shaft. As such, the fan is coupled with the LP shaft via a speed reduction gearbox, e.g., in an indirect-drive or geared-drive configuration. In this regard, the LP turbine is drivingly coupled with the LP compressor and components of the fan section. After driving each of the LP and HP turbines, the combustion products exit the turbomachine through a turbomachine exhaust nozzle. Accordingly, the turbomachine defines a working gas flow-path or core duct that extends between the core inlet and the turbomachine exhaust nozzle.

The fan section generally includes a fan having an open rotor or un-ducted fan. Further, the fan typically includes an array of fan blades secured to a hub and rotatable about a longitudinal axis. However, such fans or similar compression turbomachinery is often challenged to achieve efficiency requirements across wider ranges of operation, while maintaining acceptable stall margin.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary

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skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. 1 is a schematic, cross-sectional view of a portion of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure.

FIG. 2 is a detailed, schematic view of a portion of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating an airflow diverting assembly downstream of a fan section of the gas turbine engine.

FIG. 3 is a front view of a portion of a fluid passageway of an airflow diverting assembly of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating the fluid passageway having a constant height around a circumference thereof.

FIG. 4 is a front view of a portion of a fluid passageway of an airflow diverting assembly of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating the fluid passageway having a variable geometry around a circumference thereof.

FIG. 5 is another detailed, schematic view of a portion of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating an airflow diverting assembly downstream of a fan section of the gas turbine engine.

FIG. 6 is a front view of a portion of a rotor of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating a rotor diameter of the rotor.

FIG. 7 is a schematic view of a portion of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure, particularly illustrating an airflow path of an airflow diverting assembly.

FIG. 8 illustrates a block diagram of a controller in accordance with an exemplary embodiment of the present disclosure.

FIG. 9 illustrates a flow diagram of a method for reducing a boundary layer of an open fan of a fan section of an engine in accordance with an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.

The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative speeds within an engine unless otherwise specified. For example, a “low turbine” or “low speed turbine” defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a “high turbine” or “high speed turbine” of the engine.

The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the gas turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the gas turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the gas turbine engine.

The terms “coupled”, “fixed”, “attached to”, and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

As used herein, the terms “first”, “second”, “third” and so on may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the embodiments as they are oriented in the drawing figures. However, it is to be understood that the embodiments may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the disclosure. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.

A “third stream” as used herein means a non-primary air stream capable of increasing fluid energy to produce a minority of total propulsion system thrust. The third stream may generally receive inlet air (air from a ducted passage downstream of a primary fan) instead of freestream air (as the primary fan would). A pressure ratio of the third stream may be higher than that of the primary propulsion stream (e.g., a bypass or propeller driven propulsion stream). The thrust may be produced through a dedicated nozzle or through mixing of an airflow through the third stream with a primary propulsion stream or a core air stream, e.g., into a common nozzle.

In certain exemplary embodiments an operating temperature of the airflow through the third stream may be less than a maximum compressor discharge temperature for the engine, and more specifically may be less than 350 degrees Fahrenheit (such as less than 300 degrees Fahrenheit, such as less than 250 degrees Fahrenheit, such as less than 200 degrees Fahrenheit, and at least as great as an ambient temperature). In certain exemplary embodiments these operating temperatures may facilitate heat transfer to or from the airflow through the third stream and a separate fluid stream. Further, in certain exemplary embodiments, the airflow through the third stream may contribute less than 50% of the total engine thrust (and at least, e.g., 2% of the total engine thrust) at a takeoff condition, or more particularly while operating at a rated takeoff power at sea level, static flight speed, 86 degree fahrenheit ambient temperature operating conditions.

Furthermore in certain exemplary embodiments, aspects of the airflow through the third stream (e.g., airstream, mixing, or exhaust properties), and thereby the aforementioned exemplary percent contribution to total thrust, may passively adjust during engine operation or be modified purposefully through use of engine control features (such as fuel flow, electric machine power, variable stators, variable inlet guide vanes (IGVs), valves, variable exhaust geometry, or fluidic features) to adjust or optimize overall system performance across a broad range of potential operating conditions.

Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 10, 15, or 20 percent margin. These approximating margins may apply to a single value, either or both endpoints defining numerical ranges, and/or the margin for ranges between endpoints.

Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

The present disclosure is generally directed to a turbofan engine having a turbomachine and a rotor assembly, or fan section, with a single stage of unducted rotor blades mounted to a hub or spinner. Further, the turbomachine

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generally includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. The fan section generally includes a fan having an open rotor or un-ducted fan. Further, the fan typically includes an array of fan blades secured to the hub and rotatable about a longitudinal axis. As such, for un-ducted fans to operate efficiently at desirable flight speeds, a large hub or spinner is required, which is unique versus ducted fans having inlets that reduce the flow velocity into the fan versus flight speed. In contrast, un-ducted fans do not have the ability to reduce the flow velocity in this matter, so un-ducted fans must resolve incoming flow at higher speeds via a larger rotor. However, the increased size of the hub causes a viscous boundary layer of weak flow to develop along the hub prior to entering an engine inlet of the ducted turbomachine. This hub flow path boundary layer is a distortion that needs to be minimized to improve efficiency and stall margin of the turbofan engine. As used herein, "distortion" is generally defined as a variation in the radial velocity profile of flow entering the inlet. A corresponding definition is the variation in the radial total pressure profile. Circumferential distortion can also exist when the engine is at non-zero angle of attack to flight path. As weak flow is removed from the hub, the remaining inlet flow profile becomes more uniform.

Accordingly, the present disclosure is directed to an engine and methods for reducing boundary layer induced distortion, the engine having an airflow diverting assembly downstream of the fan section and upstream of the engine inlet that removes this boundary layer in part or in total, by means of active or passive flow suction, to reduce the distortion to be resolved by the downstream turbomachine. In an embodiment, for example, the airflow diverting assembly includes at least one fluid passageway formed into at least a portion of the hub for diverting at least a portion of the boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the engine. Thus, in an embodiment, the bled or suctioned air from the boundary layer can be used for internal component cooling (e.g., electronics, gearbox, downstream full authority digital engine control (FADEC), and/or other systems, etc.), if desired, before joining other existing ventilation flows. As such, reducing this boundary layer makes the radial aerodynamic profile entering the turbomachine more uniform. The more uniform the flow, the less constrained the immediately downstream stator vanes and rotor blades become. Lower acoustic source opportunities via either achieving same core/mid-fan flows at lower rotation speeds translates to less losses, lower fan speeds, and reduced self and interaction fan noise sources. Moreover, by removing at least some of the hub boundary layer, unsteadiness from the boundary layer into the mid-fan stream can also be reduced. Turbulence intensity informs source radiation strength both on mid-fan inlet guide vanes as well as core stream inlet noise sources.

Accordingly, the present disclosure provides many advantages over the prior art. In particular, in certain embodiments, the airflow diverting assembly of the present disclosure takes advantage of an existing gap between the rotatable hub and an engine inlet and has minimal impact to existing flow path without suction. Moreover, in an embodiment, the airflow diverting assembly of the present disclosure is configured to reduce or eliminate thick boundary layer formation along the larger hub in an un-ducted fan. Accordingly, IGV, mid-fan rotor and variable IGV design constraints (specifically for the mid-fan's IGVs upstream of the mid-fan rotor) for distortion and stall margin can be

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eased, thereby improving design flexibility of the engine. In other embodiments, the airflow diverting assembly of the present disclosure can also be implemented with engines without a mid-fan, such as a turboprop engine. In certain embodiments, the airflow diverting assembly of the present disclosure provides an opportunity for performance improvement or envelope expansion. Moreover, the airflow diverting assembly of the present disclosure is configured to improve operation of the engine for high angle of attack (AoA).

Referring now to FIG. 1, a schematic cross-sectional view of a gas turbine engine 100 ("engine 100") is provided according to an example embodiment of the present disclosure. Particularly, FIG. 1 provides a turbofan engine having a rotor assembly with a single stage of unducted rotor blades. In such a manner, the rotor assembly may be referred to herein as an "unducted fan," or the entire engine 100 may be referred to as an "unducted turbofan engine." In addition, the engine 100 of FIG. 1 includes a third stream extending from the compressor section to a rotor assembly flowpath over the turbomachine, as will be explained in more detail below. In further embodiments, the engine 100 may be a turboprop engine.

For reference, the engine 100 defines an axial direction A, a radial direction R, and a circumferential direction C. Moreover, the engine 100 defines an axial centerline or longitudinal axis 112 that extends along the axial direction A. In general, the axial direction A extends parallel to the longitudinal axis 112, the radial direction R extends outward from and inward to the longitudinal axis 112 in a direction orthogonal to the axial direction A, and the circumferential direction C extends three hundred sixty degrees (360°) around the longitudinal axis 112. The engine 100 extends between a forward end 114 and an aft end 116, e.g., along the axial direction A.

The engine 100 includes a turbomachine 120 and the rotor assembly, also referred to as a fan section 150, positioned upstream thereof. Generally, the turbomachine 120 includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. Particularly, as shown in FIG. 1, the turbomachine 120 includes a core cowl 122 that defines an annular core inlet 124. The core cowl 122 further encloses at least in part a low pressure system and a high pressure system. For example, the core cowl 122 depicted encloses and supports at least in part a booster or low pressure ("LP") compressor 126 for pressurizing the air that enters the turbomachine 120 through core inlet 124. A high pressure ("HP"), multi-stage, axial-flow compressor 128 receives pressurized air from the LP compressor 126 and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor 130 of the combustion section where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

It will be appreciated that as used herein, the terms "high/low speed" and "high/low pressure" are used with respect to the high pressure/high speed system and low pressure/low speed system interchangeably. Further, it will be appreciated that the terms "high" and "low" are used in this same context to distinguish the two systems, and are not meant to imply any absolute speed and/or pressure values.

The high energy combustion products flow from the combustor 130 downstream to an HP turbine 132. The HP turbine 132 drives the HP compressor 128 through an HP shaft 136. In this regard, the HP turbine 132 is drivingly coupled with the HP compressor 128. The high energy combustion products then flow to an LP turbine 134. The LP

turbine **134** drives the LP compressor **126** and components of the fan section **150** through an LP shaft **138**. In this regard, the LP turbine **134** is drivingly coupled with the LP compressor **126** and components of the fan section **150**. The LP shaft **138** is coaxial with the HP shaft **136** in this example embodiment. After driving each of the turbines **132**, **134**, the combustion products exit the turbomachine **120** through a turbomachine exhaust nozzle **140**.

Accordingly, the turbomachine **120** defines a working gas flowpath or core duct **142** that extends between the core inlet **124** and the turbomachine exhaust nozzle **140**. The core duct **142** is an annular duct positioned generally inward of the core cowl **122** along the radial direction R. The core duct **142** (e.g., the working gas flowpath through the turbomachine **120**) may be referred to as a second stream.

The fan section **150** includes an open fan **152**, which is the primary fan in this example embodiment. For the depicted embodiment of FIG. 1, the open fan **152** ("fan **152**") is an open rotor or unducted fan **152**. In such a manner, the engine **100** may be referred to as an open rotor engine.

As depicted, the fan **152** includes a rotor **151** having a plurality of fan blades **154** (only one shown in FIG. 1) mounted to a hub **153** or spinner that is rotatable about the longitudinal axis **112**/centerline. The fan blades **154** are rotatable, e.g., about the longitudinal axis **112**. As noted above, the fan **152** is drivingly coupled with the LP turbine **134** via the LP shaft **138**. For the embodiments shown in FIG. 1, the fan **152** is coupled with the LP shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or geared-drive configuration.

Moreover, the plurality of fan blades **154** can be arranged in equal spacing around the longitudinal axis **112**. Each fan blade **154** has a root and a tip and a span defined therebetween. Each fan blade **154** defines a central blade axis **156**. For this embodiment, each fan blade **154** of the fan **152** is rotatable about its central blade axis **156**, e.g., in unison with one another. One or more actuators **158** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan blades **154** about their respective central blades' axes **156**.

The fan section **150** further includes a fan guide vane array **160** that includes fan guide vanes **162** (only one shown in FIG. 1) disposed around the longitudinal axis **112**. For this embodiment, the fan guide vanes **162** are not rotatable about the longitudinal axis **112**. Each fan guide vane **162** has a root and a tip and a span defined therebetween. The fan guide vanes **162** may be unshrouded as shown in FIG. 1 or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **162** along the radial direction R or attached to the fan guide vanes **162**.

Each fan guide vane **162** defines a central blade axis **164**. For this embodiment, each fan guide vane **162** of the fan guide vane array **160** is rotatable about its respective central blade axis **164**, e.g., in unison with one another. One or more actuators **166** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan guide vane **162** about its respective central blade axis **164**. However, in other embodiments, each fan guide vane **162** may be fixed or unable to be pitched about its central blade axis **164**. The fan guide vanes **162** are mounted to a fan cowl **170**.

As shown in FIG. 1, in addition to the fan **152**, which is unducted, a ducted fan **184** (also referred to herein as a mid-fan) is included aft of the fan **152**, such that the engine **100** includes both a ducted and an unducted fan which both serve to generate thrust through the movement of air without passage through at least a portion of the turbomachine **120**

(e.g., without passage through the HP compressor **128** and the combustion section for the embodiment depicted). The ducted fan **184** is rotatable about the same axis (e.g., the longitudinal axis **112**) as the fan blade **154**. The ducted fan **184** is, for the embodiment depicted, driven by the LP turbine **134** (e.g., coupled to the LP shaft **138**). In the embodiment depicted, as noted above, the fan **152** may be referred to as the primary fan, and the ducted fan **184** may be referred to as a secondary fan. It will be appreciated that these terms "primary" and "secondary" are terms of convenience, and do not imply any particular importance, power, or the like.

The ducted fan **184** includes a plurality of fan blades (not separately labeled in FIG. 1) arranged in a single stage, such that the ducted fan **184** may be referred to as a single stage fan. The fan blades of the ducted fan **184** can be arranged in equal spacing around the longitudinal axis **112**. Each blade of the ducted fan **184** has a root and a tip and a span defined therebetween.

The fan cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of at least a portion of the core cowl **122** along the radial direction R. Particularly, a downstream section of the fan cowl **170** extends over a forward portion of the core cowl **122** to define a fan duct flowpath, or simply a fan duct **172**. According to this embodiment, the fan flowpath or fan duct **172** may be understood as forming at least a portion of the third stream of the engine **100**.

Incoming air may enter through the fan duct **172** through a fan duct inlet **176** and may exit through a fan exhaust nozzle **178** to produce propulsive thrust. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** along the radial direction R. Moreover, as shown, the fan duct **172** may include an array of mid-fan outlet guide vanes **187**.

The fan cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, circumferentially-spaced stationary struts **174** (only one shown in FIG. 1). The stationary struts **174** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to the stationary struts **174** may be used to connect and support the fan cowl **170** and/or core cowl **122**. In many embodiments, the fan duct **172** and the core duct **142** may at least partially co-extend (generally axially) on opposite sides (e.g., opposite radial sides) of the core cowl **122**. For example, the fan duct **172** and the core duct **142** may each extend directly from a leading edge **144** of the core cowl **122** and may partially co-extend generally axially on opposite radial sides of the core cowl **122**.

The engine **100** also defines or includes an inlet duct **180**. The inlet duct **180** extends between an engine inlet **182** and the core inlet **124**/fan duct inlet **176**. The engine inlet **182** is defined generally at the forward end of the fan cowl **170** and is positioned between the fan **152** and the fan guide vane array **160** along the axial direction A. The inlet duct **180** is an annular duct that is positioned inward of the fan cowl **170** along the radial direction R. Air flowing downstream along the inlet duct **180** is split, not necessarily evenly, into the core duct **142** and the fan duct **172** by a fan duct splitter or leading edge **144** of the core cowl **122**. In the embodiment depicted, the inlet duct **180** is wider than the core duct **142** along the radial direction R. The inlet duct **180** is also wider than the fan duct **172** along the radial direction R.

Notably, for the embodiment depicted, the engine **100** includes one or more features to increase an efficiency of a third stream thrust, $F_{N_{3S}}$ (e.g., a thrust generated by an airflow through the fan duct **172** exiting through the fan

exhaust nozzle **178**, generated at least in part by the ducted fan **184**). In particular, the engine **100** further includes an array of inlet guide vanes **186** and a strut **185** downstream of the inlet guide vanes **186**. Further, as shown, the inlet guide vanes **186** are positioned in the inlet duct **180** upstream of the ducted fan **184** and downstream of the engine inlet **182**. The array of inlet guide vanes **186** are arranged around the longitudinal axis **112**. For this embodiment, the inlet guide vanes **186** are not rotatable about the longitudinal axis **112**. Each inlet guide vane **186** defines a central blade axis (not labeled for clarity), and is rotatable about its respective central blade axis, e.g., in unison with one another. In such a manner, the inlet guide vanes **186** may be considered a variable geometry component. In an embodiment, one or more actuators may be provided to facilitate such rotation and therefore may be used to change a pitch of the inlet guide vanes **186** about their respective central blade axes. However, in other embodiments, each inlet guide vane **186** may be fixed or unable to be pitched about its central blade axis.

Further, it will be appreciated that for the embodiment depicted, the fan exhaust nozzle **178** of the fan duct **172** is further configured as a variable geometry exhaust nozzle. In such a manner, the engine **100** may include one or more actuators (not shown) for modulating the variable geometry exhaust nozzle. For example, the variable geometry exhaust nozzle may be configured to vary a total cross-sectional area (e.g., an area of the nozzle in a plane perpendicular to the longitudinal axis **112**) to modulate an amount of thrust generated based on one or more engine operating conditions (e.g., temperature, pressure, mass flowrate, etc. of an airflow through the fan duct **172**). A fixed geometry exhaust nozzle may also be adopted.

The combination of the array of inlet guide vanes **186** located upstream of the ducted fan **184**, the mid-fan outlet guide vanes **187** located downstream of the ducted fan **184**, and the fan exhaust nozzle **178** may result in a more efficient generation of third stream thrust, $F_{n_{3S}}$, during one or more engine operating conditions. Further, by introducing a variability in the geometry of the inlet guide vanes **186** and the fan exhaust nozzle **178**, the engine **100** may be capable of generating more efficient third stream thrust, $F_{n_{3S}}$, across a relatively wide array of engine operating conditions, including takeoff and climb (where a maximum total engine thrust $F_{n_{Total}}$ is generally needed) as well as cruise (where a lesser amount of total engine thrust, $F_{n_{Total}}$, is generally needed).

Moreover, referring still to FIG. 1, in exemplary embodiments, air passing through the fan duct **172** may be relatively cooler (e.g., lower temperature) than one or more fluids utilized in the turbomachine **120**. In this way, one or more heat exchangers **168** may be positioned in thermal communication with the fan duct **172**. For example, one or more heat exchangers **168** may be disposed within the fan duct **172** on one or more of the struts **174** and utilized to cool one or more fluids from the core engine with the air passing through the fan duct **172**, as a resource for removing heat from a fluid, e.g., compressor bleed air, oil or fuel.

Although not depicted, the heat exchanger **168** may be an annular heat exchanger extending substantially 360 degrees in the fan duct **172** (e.g., at least 300 degrees, such as at least 330 degrees). In such a manner, the heat exchanger **168** may effectively utilize the air passing through the fan duct **172** to cool one or more systems of the engine **100** (e.g., lubrication oil systems, compressor bleed air, electrical components, etc.). The heat exchanger **168** uses the air passing through duct **172** as a heat sink and correspondingly increases the

temperature of the air downstream of the heat exchanger **168** and exiting the fan exhaust nozzle **178**.

Referring now to FIG. 2, a close-up, schematic view of a portion of the fan section **150** of FIG. 1 is provided. Additionally, for the embodiment depicted, the engine **100** includes an airflow diverting assembly **200** downstream of the fan section **150** and upstream of the engine inlet **182**. Furthermore, in an embodiment, as shown, the airflow diverting assembly **200** generally includes at least one fluid passageway **202** formed into at least a portion of the hub **153** for diverting at least a portion **206** ("airflow **206**") of a boundary layer (as represented by arrow **204**) on a surface **157** of the hub **153** away from the surface **157** of the hub **153** to reduce distortion caused by the boundary layer **204** in the turbomachine **120** (FIG. 1). More specifically, according to the illustrated embodiment, an inlet **203** of the fluid passageway **202** may be positioned on a top side of the hub **153** at a location upstream of the engine inlet **182**. Further, as shown, the inlet **203** is fluidly coupled with the fluid passageway **202** that extends through the hub **153**. In this manner, the fluid passageway **202** receives airflow (indicated by arrows **204**) via the inlet **203** and diverts the airflow away from the surface **157** of the hub **153**.

In an embodiment, as shown, the airflow diverting assembly **200** may be a passive system that is passively controlled. As used herein, a passive system generally refers to a system that does not have variable geometry requiring valves or actuators, or electronics control loop/logic. An exemplary passive system is a series of ducts that enable the bled boundary layer (portion **206** of the boundary layer **204**) to be fed into a downstream flow stream that is of lower static pressure than the bled flow. This system has a positive pressure gradient in that the downstream flow suction the bled flow by means of eductor/ejector physics. In such embodiments, the momentum of the bled flow alone may not be enough to travel to an exhaust. As such, the third stream duct or core compartment are two "sink" locations that could educt this bled boundary layer flow. A passive control device could be a spring-loaded door—air pressure on the door causes the spring to react and the door to open or close.

In alternative embodiments, the airflow diverting assembly **200** may be an active system having various features for actively controlling the airflow **206** such as e.g., via a controller **300** (FIG. 8) that controls one or more electronics **228** described herein (FIG. 2), which may include, for example, at least one of one or more fans **213** (FIG. 4), one or more actuators **214** (FIG. 4), one or more valves **216** (FIG. 4), or similar to divert at least a portion **206** of the boundary layer flow **204** as desired. For example, in an embodiment, the airflow diverting assembly **200** may include one or more fans **213** in fluid communication with the fluid passageway **202** for increasing a pressure or velocity of the airflow **206**. In addition, the inlet **203** may include a valve for controlling the airflow **206** through the fluid passageway **202**. In particular embodiments, as an example, the fan(s) **213** may be an electric fan positioned within the fluid passageway **202** and the valve(s) described herein may be a variable throughput valve configured for controlling the flow rate of the airflow **206**.

Additionally, the airflow diverting assembly **200** may include one or more sensors **248** positioned in or adjacent to the fluid passageway **202** for determining one or more of a pressure or velocity of the airflow **206** through the fluid passageway **202**. Notably, the engine **100** may further include additional sensors for measuring parameters of the engine **100**, which may also be used to control the airflow diverting assembly **200**. Further, for the embodiment

depicted, the airflow diverting assembly **200** additionally includes the controller **300**, which may in certain exemplary embodiments be configured as part of a main aircraft controller or the engine **100**, operably connected to the fan(s) **213**, the actuator(s) **214**, the valve(s) **216**, and the sensors **248** through one or more wired or wireless communication buses. Accordingly, in certain embodiments, the controller **300** may be configured to operate the fan(s) **213**, the actuator(s) **214**, and/or the valve(s) **216** at least in part in response to the pressures and/or velocities measured by the sensors **248**.

Referring now to FIGS. 3-7, different embodiments of the fluid passageway **202** according to the present disclosure are provided. In particular, FIG. 3 illustrates a front view of an embodiment of the fluid passageway **202** according to the present disclosure. Thus, as shown in the illustrated embodiment, an inlet height H_1 of the fluid passageway **202** is constant around a circumference C of the fluid passageway **202**. In another embodiment, the inlet height H_1 of the fluid passageway **202** may vary around the circumference C of the fluid passageway **202**, e.g., so as to change a flow rate at various clockwise positions. Moreover, as shown, the fluid passageway extends continuously in a circumferential direction from 0° to 360° . As used herein, extending “continuously” in the circumferential direction from 0° to 360° means that the fluid passageway **202** is unobstructed or uninterrupted from 0° to 360° .

Accordingly, in certain embodiments, the fluid passageway **202** may extend circumferentially along any suitable angle, such as circumferentially from or between 0° and 360° starting at any clockwise location from top-dead-center of the engine **100**. For example, as shown in FIG. 3, the fluid passageway **202** extends continuously circumferentially from 0° to 360° starting at any clockwise location from top-dead-center **215** of the engine **100** around the entirety of the circumference C .

In alternative embodiments, as shown in FIG. 4, the fluid passageway **202** may extend less than the full 360° circumference and/or may have a variable geometry around the circumference C . For example, as shown, the fluid passageway **202** may be interrupted or divided into a plurality of regions **208**, **210** via at least one split member **212**. In particular, as shown, the fluid passageway **202** includes four split members **212** dividing the fluid passageway **202** into four quadrants. In other embodiments, the fluid passageway **202** may include any number of split members **212** including more than four or less than four split members to divide the fluid passageway **202** into any suitable number of regions. Additionally, the split members **212** may be substantially evenly spaced to divide the fluid passageway **202** into regions of substantially the same size, or alternatively, may vary in spacing in order to control the airflow **206** as desired.

Accordingly, as shown, one or more of the regions of the circumference C of the fluid passageway **202** may be closed (as indicated by regions **208**), whereas one or more other regions of the circumference C may be opened (as indicated by regions **210**). In such embodiments, the variable geometry of the fluid passageway **202** may be actively controlled based on a desired flow rate at different circumferential locations of the fluid passageway **202**. In other words, the different regions **208**, **210** of the circumference C of the fluid passageway **202** may be actively controlled such that the regions **208**, **210** are opened and/or closed to achieve a desired flow rate through the fluid passageway **202**. For example, in an embodiment, one or more of the plurality of

regions **208**, **210** may be actively opened or closed via one of the one or more actuators **214** and/or the one or more valves **216**.

Referring particularly to FIGS. 5 and 6, the inlet height H_1 (FIG. 5) of the fluid passageway **202** may range from about 0.1% to about 5% of a rotor diameter DR of the rotor **151** (FIG. 6). In another embodiment, as shown in FIG. 5, the inlet height H_1 of the fluid passageway **202** may range from about 1% to about 150% of a height H_2 of the engine inlet **182**. In still another embodiment, the fluid passageway **202** is configured to provide a design flow rate of about 0.1% to about 10% of a non-suction inlet flow rate into the engine inlet **182**. In further embodiments, the fluid passageway **202** is configured to provide a design flow rate of about 0.1% to about 50% of an engine core flow rate.

In additional embodiments, as shown in FIG. 5, an inner diameter wall **218** of the fluid passageway **202** may be secured to the hub **153** and is thus rotatable, whereas an outer diameter wall **220** of the fluid passageway **202** may be secured to the engine inlet **182** and is thus non-rotatable. In alternative embodiments, the inner diameter wall **218** of the fluid passageway **202** may be non-rotatable, whereas the outer diameter wall **220** of the fluid passageway **202** may be secured to the engine inlet **182** and is also non-rotatable.

In yet another embodiment, a lip **222** of the fluid passageway **202** is at least partially hidden by the hub **153** as viewed from a forward aft view of the hub **153**. In further embodiments, the lip **222** of the fluid passageway **202** may be completely hidden by the hub **153** as viewed from the forward aft view of the hub **153**. In still another embodiment, the lip **222** of the fluid passageway **202** may not be hidden by the hub **153** as viewed from the forward aft view of the hub **153**. For example, as shown in FIG. 6 the inlet height H_1 (FIG. 5) of the lip **222** may be designed such that an outer radius defined by the lip **222** is less than a maximum radius of the hub **153** upstream of the fan blades **154**. In other words, the lip **222** is radially inward of the maximum radius of the hub **153** upstream of the fan blades **154** so as to “hide” the lip **222**. Thus, by hiding the lip **222** in this manner, the inlet **203** of the fluid passageway **202** is protected from ingestion of foreign objects or debris.

In additional embodiments, as shown in FIG. 5, the fluid passageway **202** comprises one or more de-icing devices **224**. In particular, as shown, the de-icing devices **224** may include one or more de-icing blankets **224**. In such embodiments, the de-icing blankets **224** may also be used for de-icing other components of the engine **100**, such as e.g., the inlet guide vanes **186**.

Referring particularly now to FIG. 7, the portion **206** of the boundary layer **204** diverted away from the surface **157** of the hub **153** may be diverted to certain locations to cool one or more electronic or mechanical components of the engine **100** (e.g., such as electronics **228** and/or a reduction gearbox **230** of the engine **100**). In such embodiments, the electronics **228** may be cooled directly by air flowing over the components thereof. Further, in an embodiment, the reduction gearbox **230** may be cooled by air directly flowing over the reduction gearbox **230**. In further embodiments, the reduction gearbox **230** or other system oil may be cooled by means of a heat exchanger (not shown). Moreover, as shown, the portion **206** of the boundary layer **204** diverted away from the surface **157** of the hub **153** may be recirculated within at least a portion of a hub compartment **226**, a mid-fan compartment **232**, a core compartment **234**, or a pylon compartment **236** before being exhausted via one or more ventilation paths **238**, **240**, **242**, **244** that act as flow sinks, thereby enabling a passive system. In such embodi-

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ments, the engine **100** may further include one or more flow guides **246** arranged with the hub compartment **226**, the mid-fan compartment **232**, the core compartment **234**, and/or the pylon compartment **236** to direct flow to or away from the one or more electronic or mechanical components of the engine **100** as desired. In such embodiments, the flow guide(s) **246** are configured to direct flow to and/or away from specific systems in the engine **100** for thermal control benefits and to maximize aerothermal efficiency; i.e., maximum thrust recovery of the bled air to minimize overall engine specific fuel consumption (SFC).

Referring now to FIG. **8**, it should be understood that the controller **300** may be any suitable controller. For example, in one or more exemplary embodiments, the controller **300** may be a stand-alone controller, or alternatively, may be integrated into one or more of a controller for the engine **100** (FIG. **1**) or a controller for an aircraft including the engine **100**.

Referring particularly to the operation of the controller **300**, in at least certain embodiments, the controller **300** can include one or more computing device(s) **310**. The computing device(s) **310** can include one or more processor(s) **310A** and one or more memory device(s) **310B**. The one or more processor(s) **310A** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **310B** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

The one or more memory device(s) **310B** can store information accessible by the one or more processor(s) **310A**, including computer-readable instructions **310C** that can be executed by the one or more processor(s) **310A**. The instructions **310C** can be any set of instructions that when executed by the one or more processor(s) **310A**, cause the one or more processor(s) **310A** to perform operations. In some embodiments, the instructions **310C** can be executed by the one or more processor(s) **310A** to cause the one or more processor(s) **310A** to perform operations, such as any of the operations and functions for which the controller **300** and/or the computing device(s) **310** are configured, the operations for operating the engine **100**, as described herein, and/or any other operations or functions of the one or more computing device(s) **310**. The instructions **310C** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **310C** can be executed in logically and/or virtually separate threads on the one or more processor(s) **310A**. The one or more memory device(s) **310B** can further store data **310D** that can be accessed by the one or more processor(s) **310A**. For example, the data **310D** can include data indicative of power flows, data indicative of engine/aircraft operating conditions, and/or any other data and/or information described herein.

The computing device(s) **310** can also include a network interface **310E** used to communicate, for example, with the other components of the engine **100** (e.g., via a communication network). The network interface **310E** can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components. One or more devices can be configured to receive one or more commands from the computing device(s) **310** or provide one or more commands to the computing device(s) **310**.

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The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

Referring now to FIG. **9**, a flow diagram of an embodiment of a method **400** for reducing a boundary layer of an open fan of a fan section of an engine is illustrated. In general, the method **400** is described herein as relating to the engine **100** (FIG. **1**) and airflow diverting assembly **200** (FIG. **2**). However, it should be appreciated that the disclosed method **400** may be implemented using any other suitable engine and airflow diverting assembly now known or later developed in the art. In addition, although FIG. **9** depicts steps performed in a particular order for purposes of illustration and discussion, the methods described herein are not limited to any particular order or arrangement. One skilled in the art, using the disclosures provided herein, will appreciate that various steps of the methods can be omitted, rearranged, combined and/or adapted in various ways.

As shown at (**402**), the method **400** includes forming an airflow diverting assembly downstream of an open fan and upstream of an engine inlet of a turbomachine of an engine, the airflow diverting assembly comprising at least one fluid passageway formed into at least a portion of a hub of the open fan. As shown at (**404**), the method **400** includes diverting, via the at least one fluid passageway, at least a portion of a boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the turbomachine.

As will be appreciated from the description herein, various embodiments of a gas turbine engine are provided. Certain of these embodiments may be an unducted, single rotor gas turbine engine. Various additional aspects of one or more of these embodiments are discussed below. These exemplary aspects may be combined with one or more of the exemplary gas turbine engine(s) discussed above with respect to the figures.

For example, in some embodiments of the present disclosure, the engine may include a heat exchanger located in an annular duct, such as in a third stream. The heat exchanger may extend substantially continuously in a circumferential direction of the gas turbine engine (e.g., at least about 300 degrees, such as at least about 330 degrees).

In one or more of these embodiments, a threshold power or disk loading for a fan (e.g., an unducted single rotor or primary forward fan) may range from 25 horsepower per square foot (hp/ft²) or greater at cruise altitude during a cruise operating mode. In particular embodiments of the engine, structures and methods provided herein generate power loading between 80 hp/ft² and 160 hp/ft² or higher at cruise altitude during a cruise operating mode.

In various embodiments, an engine of the present disclosure is applied to a vehicle with a cruise altitude up to approximately 65,000 ft. In certain embodiments, cruise altitude is between approximately 28,000 ft and approximately 45,000 ft. In still certain embodiments, cruise altitude is expressed in flight levels based on a standard air pressure at sea level, in which a cruise flight condition is

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between FL280 and FL650. In another embodiment, cruise flight condition is between FL280 and FL450. In still certain embodiments, cruise altitude is defined based at least on a barometric pressure, in which cruise altitude is between approximately 4.85 psia and approximately 0.82 psia based on a sea level pressure of approximately 14.70 psia and sea level temperature at approximately 59 degrees Fahrenheit. In another embodiment, cruise altitude is between approximately 4.85 psia and approximately 2.14 psia. It should be appreciated that in certain embodiments, the ranges of cruise altitude defined by pressure may be adjusted based on a different reference sea level pressure and/or sea level temperature.

As such, it will be appreciated that an engine of such a configuration may be configured to generate at least about 25,000 pounds and less than about 80,000 of thrust during operation at a rated speed, such as between about 25,000 and 50,000 pounds of thrust during operation at a rated speed, such as between about 25,000 and 40,000 pounds of thrust during operation at a rated speed.

In various exemplary embodiments, the fan may include twelve (12) fan blades. From a loading standpoint, such a blade count may allow a span of each blade to be reduced such that the overall diameter of the primary fan may also be reduced (e.g., to about twelve feet in one exemplary embodiment). That said, in other embodiments, the fan may have any suitable blade count and any suitable diameter. In certain suitable embodiments, the fan includes at least eight (8) blades. In another suitable embodiment, the fan may have at least twelve (12) blades. In yet another suitable embodiment, the fan may have at least fifteen (15) blades. In yet another suitable embodiment, the fan may have at least eighteen (18) blades. In one or more of these embodiments, the fan includes twenty-six (26) or fewer blades, such as twenty (20) or fewer blades.

Further, in certain exemplary embodiments, the rotor assembly may define a rotor diameter (or fan diameter) of at least 10 feet, such as at least 11 feet, such as at least 12 feet, such as at least 13 feet, such as at least 15 feet, such as at least 17 feet, such as up to 28 feet, such as up to 26 feet, such as up to 24 feet, such as up to 18 feet.

In various embodiments, it will be appreciated that the engine includes a ratio of a quantity of vanes to a quantity of blades that could be less than, equal to, or greater than 1:1. For example, in particular embodiments, the engine includes twelve (12) fan blades and ten (10) vanes. In other embodiments, the vane assembly includes a greater quantity of vanes to fan blades. For example, in particular embodiments, the engine includes ten (10) fan blades and twenty-three (23) vanes. For example, in certain embodiments, the engine may include a ratio of a quantity of vanes to a quantity of blades between 1:2 and 5:2. The ratio may be tuned based on a variety of factors including a size of the vanes to ensure a desired amount of swirl is removed for an airflow from the primary fan.

Additionally, in certain exemplary embodiments, where the engine includes the third stream and a mid-fan (a ducted fan aft of the primary, forward fan), a ratio $R1/R2$ may be between about 1 and 10, or 2 and 7, or at least about 3.3, at least about 3.5, at least about 4 and less than or equal to about 7, where $R1$ is the radius of the primary fan and $R2$ is the radius of the mid-fan.

It should be appreciated that various embodiments of the engine, such as the single unducted rotor engine depicted and described herein, may allow for normal subsonic aircraft cruise altitude operation at or above Mach 0.5. In certain embodiments, the engine allows for normal aircraft opera-

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tion between Mach 0.55 and Mach 0.85 at cruise altitude. In still particular embodiments, the engine allows for normal aircraft operation between Mach 0.75 and Mach 0.85. In certain embodiments, the engine allows for rotor blade tip speeds at or less than 750 feet per second (fps). In other embodiments, the rotor blade tip speed at a cruise flight condition can be 650 to 900 fps, or 700 to 800 fps.

A fan pressure ratio (FPR) for the fan of the fan assembly can be 1.04 to 1.20, or in some embodiments 1.05 to 1.1, or in some embodiments less than 1.08, as measured across the fan blades at a cruise flight condition.

In order for the gas turbine engine to operate with a fan having the above characteristics to define the above FPR, a gear assembly may be provided to reduce a rotational speed of the fan assembly relative to a driving shaft (such as a low pressure shaft coupled to a low pressure turbine). In some embodiments, a gear ratio of the input rotational speed to the output rotational speed is greater than 4:1. For example, in particular embodiments, the gear ratio is within a range of 4.1 to 14.0, within a range of 4.5 to 14.0, or within a range of 6.0 to 14.0. In certain embodiments, the gear ratio is within a range of 4.5 to 12 or within a range of 6.0 to 11.0. As such, in some embodiments, the fan can be configured to rotate at a rotational speed of 700 to 1500 rpm at a cruise flight condition, while the power turbine (e.g., the low-pressure turbine) is configured to rotate at a rotational speed of 2,500 to 15,000 rpm at a cruise flight condition. In particular embodiments, the fan can be configured to rotate at a rotational speed of 850 to 1,350 rpm at a cruise flight condition, while the power turbine is configured to rotate at a rotational speed of 5,000 to 10,000 rpm at a cruise flight condition.

With respect to a turbomachine of the gas turbine engine, the compressors and/or turbines can include various stage counts. As disclosed herein, the stage count includes the number of rotors or blade stages in a particular component (e.g., a compressor or turbine). For example, in some embodiments, a low pressure compressor may include 1 to 8 stages, a high-pressure compressor may include 8 to 15 stages, a high-pressure turbine may include 1 to 2 stages, and/or a low pressure turbine (LPT) may include 3 to 7 stages. In particular, the LPT may have 4 stages, or between 4 and 7 stages. For example, in certain embodiments, an engine may include a one stage LP compressor, an 11 stage HP compressor, a two stage HP turbine, and 4 stages, or between 4 and 7 stages for the LPT. As another example, an engine can include a three stage LP compressor, a 10 stage HP compressor, a two stage HP turbine, and a 7 stage LP turbine.

A core engine is generally encased in an outer casing defining one half of a core diameter (Dcore), which may be thought of as the maximum extent from a centerline axis (datum for R). In certain embodiments, the engine includes a length (L) from a longitudinally (or axial) forward end to a longitudinally aft end. In various embodiments, the engine defines a ratio of L/D_{core} that provides for reduced installed drag. In one embodiment, L/D_{core} is at least 2. In another embodiment, L/D_{core} is at least 2.5. In some embodiments, the L/D_{core} is less than 5, less than 4, and less than 3. In various embodiments, it should be appreciated that the L/D_{core} is for a single unducted rotor engine.

The reduced installed drag may further provide for improved efficiency, such as improved specific fuel consumption. Additionally, or alternatively, the reduced drag may provide for cruise altitude engine and aircraft operation at the above describe Mach numbers at cruise altitude. Still particular embodiments may provide such benefits with

reduced interaction noise between the blade assembly and the vane assembly and/or decreased overall noise generated by the engine by virtue of structures located in an annular duct of the engine.

Additionally, it should be appreciated that ranges of power loading and/or rotor blade tip speed may correspond to certain structures, core sizes, thrust outputs, etc., or other structures at the core engine. However, as previously stated, to the extent one or more structures provided herein may be known in the art, it should be appreciated that the present disclosure may include combinations of structures not previously known to combine, at least for reasons based in part on conflicting benefits versus losses, desired modes of operation, or other forms of teaching away in the art.

Although depicted above as an unshrouded or open rotor engine in the embodiments depicted above, it should be appreciated that aspects of the disclosure provided herein may be applied to shrouded or ducted engines, partially ducted engines, aft-fan engines, or other gas turbine engine configurations, including those for marine, industrial, or aero-propulsion systems. Certain aspects of the disclosure may be applicable to turbofan, turboprop, or turboshaft engines. However, it should be appreciated that certain aspects of the disclosure may address issues that may be particular to unshrouded or open rotor engines, such as, but not limited to, issues related to gear ratios, fan diameter, fan speed, length (L) of the engine, maximum diameter of the core engine (Dcore) of the engine, L/Dcore of the engine, desired cruise altitude, and/or desired operating cruise speed, or combinations thereof.

Further aspects are provided by the subject matter of the following clauses:

An engine, comprising: a fan section comprising an open fan defining an axial centerline and a rotor, the rotor comprising a plurality of fan blades mounted to a hub and being rotatable about the axial centerline; a turbomachine downstream of the fan section, the turbomachine comprising a core cowl that defines an engine inlet and encloses turbomachinery; and an airflow diverting assembly downstream of the fan section and upstream of the engine inlet, the airflow diverting assembly comprising at least one fluid passageway formed into at least a portion of the hub for diverting at least a portion of a boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the turbomachinery.

The engine of any preceding clause, wherein the airflow diverting assembly is a passive system that is passively controlled.

The engine of any preceding clause, wherein a height of the at least one fluid passageway is constant around a circumference of the at least one fluid passageway.

The engine of any preceding clause, wherein the at least one fluid passageway extends circumferentially between 0° and 360° starting at any clockwise location from top-dead-center of the engine.

The engine of any preceding clause, wherein the at least one fluid passageway extends continuously in a circumferential direction from 0° to 360°.

The engine of any preceding clause, wherein the airflow diverting assembly is an active system that is actively controlled.

The engine of any preceding clause, wherein the airflow diverting assembly is actively controlled via a controller.

The engine of any preceding clause, wherein the at least one fluid passageway comprises a variable geometry.

The engine of any preceding clause, wherein the variable geometry of the at least one fluid passageway is actively controlled based on a desired flow rate at different circumferential locations of the at least one fluid passageway.

The engine of any preceding clause, wherein the airflow diverting assembly further comprises at least one of one or more fans, one or more actuators, or one or more valves.

The engine of any preceding clause, wherein the at least one fluid passageway is divided into a plurality of regions via at least one split member.

The engine of any preceding clause, wherein one or more of the plurality of regions are actively opened or closed via one of the one or more actuators or the one or more valves.

The engine of any preceding clause, wherein the at least one fluid passageway defines an inlet height ranging between about 0.1% and about 5% of a rotor diameter of the rotor.

The engine of any preceding clause, wherein the at least one fluid passageway defines an inlet height ranging between about 1% and about 150% of a height of the engine inlet.

The engine of any preceding clause, wherein the at least one fluid passageway defines an inlet height ranging from about 0.1% to about 5% of a rotor diameter of the rotor.

The engine of any preceding clause, wherein the at least one fluid passageway defines an inlet height ranging from about 1% to about 150% of a height of the engine inlet.

The engine of any preceding clause, wherein an inner diameter wall of the at least one fluid passageway is secured to the hub and is thus rotatable and an outer diameter wall of the at least one fluid passageway is secured to the engine inlet and is thus non-rotatable.

The engine of any preceding clause, wherein an inner diameter wall of the at least one fluid passageway is non-rotatable and an outer diameter wall of the at least one fluid passageway is secured to the engine inlet and is thus non-rotatable.

The engine of any preceding clause, wherein a lip of the at least one fluid passageway is at least partially hidden by the hub as viewed from a forward aft view of the hub.

The engine of any preceding clause, wherein the at least one fluid passageway comprises one or more de-icing devices.

The engine of any preceding clause, wherein the portion of the boundary layer diverted away from the surface of the hub is diverted to cool one or more electronic or mechanical components of the engine.

The engine of any preceding clause, wherein the portion of the boundary layer diverted away from the surface of the hub to cool the one or more electronic or mechanical components of the engine is recirculated within at least a portion of a hub compartment, a mid-fan compartment, a core compartment, or a pylon compartment before being exhausted via one or more ventilation paths that act as flow sinks enabling a passive system.

The engine of any preceding clause, further comprising one or more flow guides arranged with at least one of the hub compartment, the mid-fan compartment, the core compartment, or the pylon compartment to direct flow to or away from the one or more electronic or mechanical components of the engine.

The engine of any preceding clause, wherein the engine is one of a turbofan engine or a turboprop engine.

A method for reducing a boundary layer of an open fan of a fan section of an engine, the method comprising: forming an airflow diverting assembly downstream of the open fan and upstream of an engine inlet of a turbomachine of the

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engine, the airflow diverting assembly comprising at least one fluid passageway formed into at least a portion of a hub of the open fan; and diverting, via the at least one fluid passageway, at least a portion of a boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the turbomachine.

The method of any preceding clause, wherein diverting, via the at least one fluid passageway, at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises: passively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub.

The method of any preceding clause, wherein diverting, via the at least one fluid passageway, at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises: actively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub.

The method of any preceding clause, wherein the at least one fluid passageway comprises a variable geometry, wherein actively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises: actively controlling the variable geometry of the at least one fluid passageway via at least one of one or more fans, one or more actuators, or one or more valves based on a desired flow rate at different circumferential locations of the at least one fluid passageway.

The method of any preceding clause, further comprising de-icing the at least one fluid passageway.

The method of any preceding clause, further comprising directing the portion of the boundary layer diverted away from the surface of the hub to one or more electronic or mechanical components of the engine for cooling.

The method of any preceding clause, wherein directing the portion of the boundary layer diverted away from the surface of the hub to one or more electronic or mechanical components of the engine for cooling further comprises: recirculating the portion of the boundary layer diverted away from the surface of the hub within at least a portion of a hub compartment, a mid-fan compartment, or a core compartment before being exhausted via one or more ventilation paths that act as flow sinks enabling a passive system.

The method of any preceding clause, further comprising arranging one or more flow guides with at least one of the hub compartment, the mid-fan compartment, or the core compartment to direct flow to or away from the one or more electronic or mechanical components of the engine.

This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent

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structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A method for reducing a boundary layer of an open fan of a fan section of an engine, the method comprising:

forming an airflow diverting assembly downstream of the open fan and upstream of an engine inlet of a turbomachine of the engine, the airflow diverting assembly comprising at least one fluid passageway formed into at least a portion of a hub of the open fan; and

diverting, via the at least one fluid passageway, at least a portion of a boundary layer on a surface of the hub away from the surface of the hub to reduce distortion caused by the boundary layer in the turbomachine.

2. The method of claim 1, wherein diverting, via the at least one fluid passageway, at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises:

passively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub.

3. The method of claim 1, wherein diverting, via the at least one fluid passageway, at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises:

actively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub.

4. The method of claim 3, wherein the at least one fluid passageway comprises a variable geometry, wherein actively controlling the airflow diverting assembly to divert at least the portion of the boundary layer on the surface of the hub away from the surface of the hub further comprises:

actively controlling the variable geometry of the at least one fluid passageway via at least one of one or more fans, one or more actuators, or one or more valves based on a desired flow rate at different circumferential locations of the at least one fluid passageway.

5. The method of claim 1, further comprising de-icing the at least one fluid passageway.

6. The method of claim 1, further comprising directing the portion of the boundary layer diverted away from the surface of the hub to one or more electronic or mechanical components of the engine for cooling.

7. The method of claim 6, wherein directing the portion of the boundary layer diverted away from the surface of the hub to one or more electronic or mechanical components of the engine for cooling further comprises:

recirculating the portion of the boundary layer diverted away from the surface of the hub within at least a portion of a hub compartment, a mid-fan compartment, or a core compartment before being exhausted via one or more ventilation paths that act as flow sinks enabling a passive system.

8. The method of claim 7, further comprising arranging one or more flow guides with at least one of the hub compartment, the mid-fan compartment, or the core compartment to direct flow to or away from the one or more electronic or mechanical components of the engine.

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